

Parham Khosh Solat

Data Analyst

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SUMMARY

Six analytics projects in two years: mapped market expansion potential across all 20 Italian regions for Fater S.p.A. by joining proprietary sales data with ISTAT census records, built a Kafka pipeline streaming 55 equities daily, and shipped three dashboards for non-technical users. MSc Data Science at Federico II. Available immediately.

TECHNICAL SKILLS

Data Analysis: Advanced SQL (CTEs, window functions), Python (Pandas, NumPy, SciPy), PostgreSQL, Microsoft Excel, Google Sheets
Statistics & Testing: Statistical Analysis, Hypothesis Testing, A/B Testing, Regression, Classification, Time-Series Analysis
Visualisation & BI: Power BI, Plotly, Seaborn, Tableau, Folium, GeoPandas, Streamlit
Machine Learning: Scikit-learn, XGBoost, LightGBM, PyTorch, HuggingFace Transformers, Statsmodels
Data Engineering: Apache Kafka, PySpark, ETL Pipelines, REST APIs, GraphQL
Tools: Docker, Git, GitHub, Jupyter, HuggingFace Spaces

PROJECTS

Retail Geospatial Analytics — Fater S.p.A. Industry Challenge

2024 | Team of 4

Python · SQL · GeoPandas · Folium · Scikit-learn · Pandas · MySQL

- Joined Fater's proprietary sales data with ISTAT sociodemographic census records across all 20 Italian NUTS-2 regions using SQL, merging two heterogeneous sources with no prior link into a single analysis-ready dataset stored in MySQL.
- Ran correlation analysis and K-means clustering to score all 20 regions by expansion potential and identify the top 3 performance drivers; findings challenged assumptions the business team held before the analysis.
- Presented results solo to Fater S.p.A. leadership using choropleth dashboards in GeoPandas and Folium, calculating store potential per capita by district for a non-technical audience; received an individual participation certificate.

Real-Time Equity Clustering Pipeline

30/30 con lode | 2024

Apache Kafka · PySpark · Scikit-learn · PCA · K-means · Plotly · Yahoo Finance API

- Built a Kafka streaming pipeline with 55 individual topics, one per ticker, to ingest 6 months of daily OHLCV data for 55 US equities (April–September 2023); processed the feed with PySpark and engineered daily-return and rolling-volatility features from raw price series.
- Selected $K=4$ clusters via both Silhouette score and Elbow analysis; labelled each archetype (growth, defensive, cyclical, high-volatility) and applied PCA for 3-D visualisation of cluster separation.
- Gave portfolio managers a concrete basis for allocation decisions: four named archetypes with distinct risk and return profiles, replacing heuristic groupings with a repeatable, evidence-backed process.

Multi-Model Time-Series Forecasting — OULAD Educational Dataset

30/30 con lode | 2025

Python · SARIMA · ARIMAX · Facebook Prophet · PyTorch · Streamlit

- Forecasted student engagement peaks across a university semester from a 400 MB click-stream dataset (Open University Learning Analytics), so academic planners could allocate resources ahead of demand rather than after.
- Benchmarked four models: SARIMA and ARIMAX with 7 engineered lag features (1, 2, 3, 7, 14, 21, 28 days), Facebook Prophet, and a custom PyTorch CNN with 30 lag features trained for 50 epochs at batch size 32. The CNN produced the best MAE and MAPE on held-out data; SARIMA was kept as the interpretable seasonal baseline for non-technical stakeholders.
- Built a Streamlit forecast dashboard so planners could query predictions and run scenarios without writing code.

Pest Population Forecasting

30/30 con lode | 2025 | Live on HuggingFace Spaces

Python · SQL · Pandas · Scikit-learn · XGBoost · LightGBM · Random Forest · ARIMAX · Streamlit

- Inherited sensor data from 5 monitoring locations combining meteorological readings with entomological catch records (245 samples, July–August 2024): inconsistent field names, mixed date formats, missing values above 40% in key columns, and a 10.67:1 class imbalance. Cleaned and standardised entirely in SQL and Pandas before any modelling began.
- Tested 6 regression models (ARIMAX, SARIMAX, Prophet, Random Forest, XGBoost, LightGBM) and 6 classification models including deep learning architectures using TimeSeriesSplit cross-validation; Random Forest topped both brackets: Test MAE 0.340 and R^2 0.353 on regression, F1-score 0.667 and AUC 0.919 on classification — beating all 5 competing models.

- Deployed the final pipeline as an interactive Streamlit app on HuggingFace Spaces, accessible without local installation.

Florence-2 Fine-Tuning for Visual Question Answering

30/30 con lode | 2024

Python · PyTorch · HuggingFace Transformers · Florence-2 · Streamlit

- Benchmarked three vision-language architectures for open-ended VQA on 150K question-image pairs from VQA v2.0 abstract scenes (60K train, 30K validation, 60K test): PaliGemma 3B ran out of memory at 40 GB, BLIP produced unstable gradients, Florence-2 at 771M parameters trained cleanly on A100 hardware.
- Fine-tuned Florence-2 with AdamW (lr 1e-5, batch size 8) over 3 epochs: training loss fell from 0.307 to 0.111, a 64% reduction; validation loss stabilised at 0.202. Traced and fixed a JSON-to-model preprocessing mismatch that had been silently corrupting accuracy in the baseline evaluation pipeline.
- Shipped a Streamlit app to HuggingFace Spaces with beam-search controls, confidence scores, and session-history export, with no backend infrastructure needed.

TalentSonar — AI Recruiting Platform

Hackathon | 2025 | Live on HuggingFace Spaces

Python · GitHub GraphQL API · Google Gemini API · Streamlit

- Built a GraphQL ingestion layer that pulls structured candidate profiles from GitHub, including repository list, primary languages, and contribution activity, for automated recruiter screening.
- Connected Google Gemini to parse job descriptions and extract required skills automatically; deployed the full application on HuggingFace Spaces with candidate browsing, a questionnaire system, and job management.

MSc Thesis — Preference Shielding in Human-Robot Interaction

In progress | 2025–2026

Python · Reinforcement Learning · JavaScript · Docker · Statistical Analysis

- Designing RL experiments to study how agents learn to respect user preferences in interactive settings, with implications for preference-aware safety in human-robot systems.
- Building the experiment platform from scratch: Python backend, JavaScript frontend, Docker containerisation, and a statistical pipeline for participant data collection and processing.

TEACHING EXPERIENCE

Teaching Assistant — Computer Architecture & Microprocessors

Amol University, Iran

Supported undergraduate instruction across 4 semesters (2 per course), covering hardware fundamentals and embedded systems.

EDUCATION

MSc in Data Science

Sept 2023 – Expected Oct 2026

University of Naples Federico II · Naples, Italy · GPA 28.3/30 · 3× 30 e lode (highest distinction)

Pre-Master Foundation Programme (NCUK)

2021 – 2022

Avicenna International College · Budapest, Hungary

BSc in Information Technology Engineering

2013 – 2017

Amol University of Special Modern Technologies · Amol, Iran

LANGUAGES

Persian (native) English (C1 — full professional proficiency)